Compositional Navigation Tensor

A hash-traceable instrument for compositional dynamics

CoDaWork 2026

P. Higgins

The instrument reads. The expert decides. The hashes hold the line.

What CNT adds on top of the CoDa toolkit

Three additions that complement the established CoDa methods:

Trajectory operators

bearings, ω, helmsman, period-2 attractor, IR class — first-class objects in the algebra, computed alongside the standard ILR.

End-to-end determinism

raw CSV → JSON → PDF byte-identical. content_sha256 + engine_signature on every page, supporting reproducibility for both pipelines.

Cross-dataset inference

Stage 4 reports compare attractor amplitude, depth tower, and IR distribution across multiple datasets in one document.

These ship alongside the classical plates — both languages, one report.

Variation matrix · biplot · balance dendrogram · SBP · ternary · scree — same engine.

CNT Pipeline -- Stage 1: Raw Data to CLR Coordinate

RAW DATA

$$Y = \{ Y_j(t) \} \; j = 1..D, \; t = 1..N$$

Absolute values (TWh, counts, mass) -- D carriers x N time steps

↓ closure

CLOSURE -- Composition x(t)

$$x_j(t) = Y_j(t) / \mathrm{Sum}_k \, Y_k(t)$$

x in S^D (open simplex) -- Sum x_j = 1 -- x_j > 0

↓ log-ratio

CLR TRANSFORM -- Higgins Coordinate h(t)

$$h_j(t) = \ln(x_j(t)) - (1/D) \, \mathrm{Sum}_k \ln(x_k(t))$$

$$= \ln(x_j / g(x)) \text{ where } g(x) = \text{geometric mean}$$

h in R^D | Sum h_j = 0 | Unit: HLR (Higgins Log-Ratio Level) | D-1 degrees of freedom

The instrument reads. The expert decides. The loop stays open.

Math step 1+2 — Closure to simplex + CLR transform

Same operation in both pipelines. CNT adds the audit metadata.

Classical CoDa

closure

$$x_j = Y_j / \mathrm{sum}(Y)$$

zero replacement

$$x_j \leftarrow \max(x_j, \delta)$$

CLR (Aitchison 1986)

$$\mathrm{clr}(x)_j = \ln(x_j) - \langle \ln x \rangle$$

δ chosen by practitioner

Foundation method, well-understood.

CNT (built on top)

closure (same)

$$x_j = Y_j / \mathrm{sum}(Y)$$

zero replacement (disclosed)

$$x_j \leftarrow \max(x_j, DEFAULT\_DELTA)$$

δ recorded in metadata.engine_config

CLR (same)

$$\mathrm{clr}(x)_j = \ln(x_j) - \langle \ln x \rangle$$

δ enters content_sha256

Same arithmetic, with audit-grade lineage.

CNT preserves the established CoDa step and adds an explicit lineage record.

Math step 3 — Helmert orthonormal ILR projection

Egozcue (2003) basis. Both pipelines use it. CNT publishes V in the JSON.

Classical CoDa

build Helmert basis V

$$V \text{ is } (D\text{-}1) \times D, \text{ orthonormal}$$

computed inline; published per workflow

project

$$ilr(x) = V \cdot clr(x)$$

three orthogonal views, per-pair scaling

flexible, well-understood

CNT (additions)

basis published in JSON

tensor.helmert_basis.coefficients

V part of the canonical output

one matmul + shared window

$$ILR\_T \times (D\text{-}1) = V \cdot CLR\_T \times D^T$$

XY, XZ, YZ are slices of the same matrix

FIXED scale across all 3 panels

CNT keeps Egozcue's projection and adds caching plus a shared display scale.

Math step 4 — Pairwise bearing θ_i,j

Both formulae compute the same angle. atan2 keeps better numerical headroom near the axes.

arccos formulation

$$\theta = \arccos(\langle u,v\rangle / (|u|\cdot|v|))$$

range: [0, π] (sign restored via sign-test)

characteristics

- ~12 float ops per pair
- noise floor √ε near 0 and π
- two ‖·‖ + dot product
- well-established formulation

atan2 formulation (CNT)

$$\theta = \mathrm{atan2}(y\_i,j \, , \, x\_i,j)$$

range: [-π, π] (full quadrant)

characteristics

- ~4 float ops per pair
- machine precision everywhere
- quadrant implicit
- single CPU instruction (IEEE 754)

3× fewer ops · ~10^7 better headroom near 0/π

Lock-detection stable to IEEE floor

arccos remains correct; atan2 gives extra numerical room for trajectory locks.

Math step 5 — Metric tensor and dual involution $M^2 = I$

A new addition CNT introduces. It licences the recursive depth tower (steps 6–7).

Classical CoDa

variation matrix is the local equivalent

log-ratio variance per carrier pair —
the established structural diagnostic.

Excellent for pairwise CoDa structure;
CNT's $M^2 = I$ is the additional
contraction certificate that licences
the recursive depth tower.

The variation matrix ships in Stage 2.

CNT addition

build metric tensor M(x)

$$M \in \mathbb{R}\text{^}(D \times D)$$

verify involution per timestep

$$M^2 = I$$

mean residual ~ 1e-15 (IEEE floor)

$M^2 = I$ gives the Banach-contraction certificate.
Required for the recursive depth tower to provably converge.

Recorded in diagnostics.higgins_extensions.involution_proof on every CNT JSON.

Math step 6+7 — Depth tower → period-2 attractor → IR class

CNT additions, built on the M² = I contraction certificate.

DEPTH TOWER

Energy and curvature towers recursively iterate the metric dual

$$\mathit{level}\ k \rightarrow \mathit{level}\ k{+}1$$

at each level: ω(k), Hs(k)

convergence:
ω(k) → noise floor

Banach satisfied → stop

depth = number of levels

PERIOD-2 ATTRACTOR

tower converges to two alternating Hs samples

c_even, c_odd

A = | c_even - c_odd |

amplitude is the headline structural diagnostic

Lyapunov, residual, period all recorded in JSON

contraction ratio < 1

IR CLASS (8)

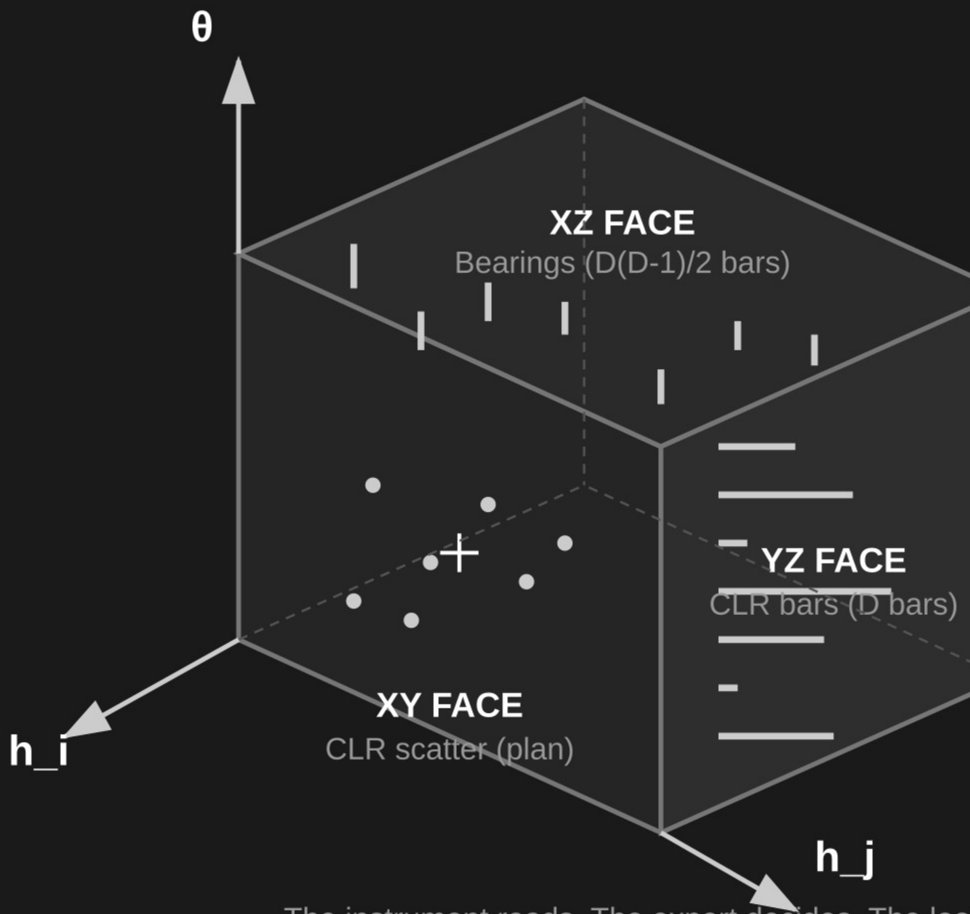
CRITICALLY_DAMPED
LIGHTLY_DAMPED
MODERATELY_DAMPED
OVERDAMPED_EXTREME
UNDAMPED
ENERGY_STABLE_FIXED
CURVATURE_VERTEX_FLAT
D2_DEGENERATE

classification by A and tower termination

Each box is one block in diagnostics.depth.higgins_extensions of every CNT JSON.
Anyone with the engine can recompute the chain and verify the IR class.

These structures sit alongside the established CoDa toolkit — additions, not replacements.

CBS Cube — Three Orthogonal Face Projections



CBS

Compositional Bearing Scope

3 faces = 3 simultaneous views

All scales FIXED across time

Unit: HLR

Origin = geometric mean

(barycentre of the simplex)

Axes:

h_i, h_j = CLR coordinates

θ = bearing angle

The instrument reads. The expert decides. The loop stays open.

One-picture summary — the full hash-traceable pipeline

raw CSV (source_file_sha256) → **cnt engine 2.0.4** (engine_signature) → **canonical JSON** (content_sha256 (schema 2.1.0)) → **4-stage atlas** (PDF + interactive HTML)

Four stages, one per dataset, one per project

Stage 1 (Order 1)
per-timestep ortho plate
ILR-Helmert triplet
T plates per dataset

Stage 2 (Order 2)
single report per dataset
geometry + dynamics
28 plates

Stage 3 (Order 3)
depth / IR / attractor
single report per dataset
11 plates

Stage 4 (Order 4+)
cross-dataset inference
group-level report
11 plates

EVERY page footer carries the full chain

engine: 2.0.4 (sig...) | data: csha... | source: ssha... | date

Anyone with the raw CSV can verify any plate in ~2 minutes.

25 reference experiments / 25 PASS / determinism gate.

The instrument reads. The expert decides. The hashes hold the line.